

Ratio & Proportion

Answer sheet · 26 marks total

Section A — Simplifying ratios

A1. Divide both by 4 → **4:5** (1)

A2. Divide both by 9 → **5:2** (1)

A3. Divide both by 5 (1) → **1:2.40** (1)

A4. Total parts = 10, blue = **7/10** (1)

Section B — Sharing in a given ratio

B1. $84 \div 7 = £12$ (1) → shares = **£24 : £60** (1)

B2. $120 \div 6 = 20$ (1) → shares = **20 : 40 : 60** (1) → check total = 120 (1)

B3. 4 parts = £48 → 1 part = £12 (1) → total (7 parts) = **£84** (1)

Section C — Direct proportion

C1. $3.50 \div 5 = £0.70$ (1) → $8 \times 0.70 =$ **£5.60** (1)

C2. $200 \div 4 = 50$ g per person (1) → $50 \times 10 =$ **500 g** (1)

C3. Shop A: $2.40 \div 6 = £0.40$ per apple (1) Shop B: $3.80 \div 10 = £0.38$ per apple (1) → **Shop B** is better value, cheaper per apple (1)

Section D — Inverse proportion

D1. Worker-hours = $4 \times 15 = 60$ (1) → $60 \div 6$ (1) → = **10 hours** (1)

D2. $k = 5 \times 12 = 60$ (1) → $y = 60 \div 3 =$ **20** (1)

D3. Distance = $40 \times 3 = 120$ miles (1) → time = $120 \div 60 =$ **2 hours** (1)

Marking note: award method marks for correctly identifying the total parts, the unitary value, or the constant of proportionality, even if the final answer contains an arithmetic error.